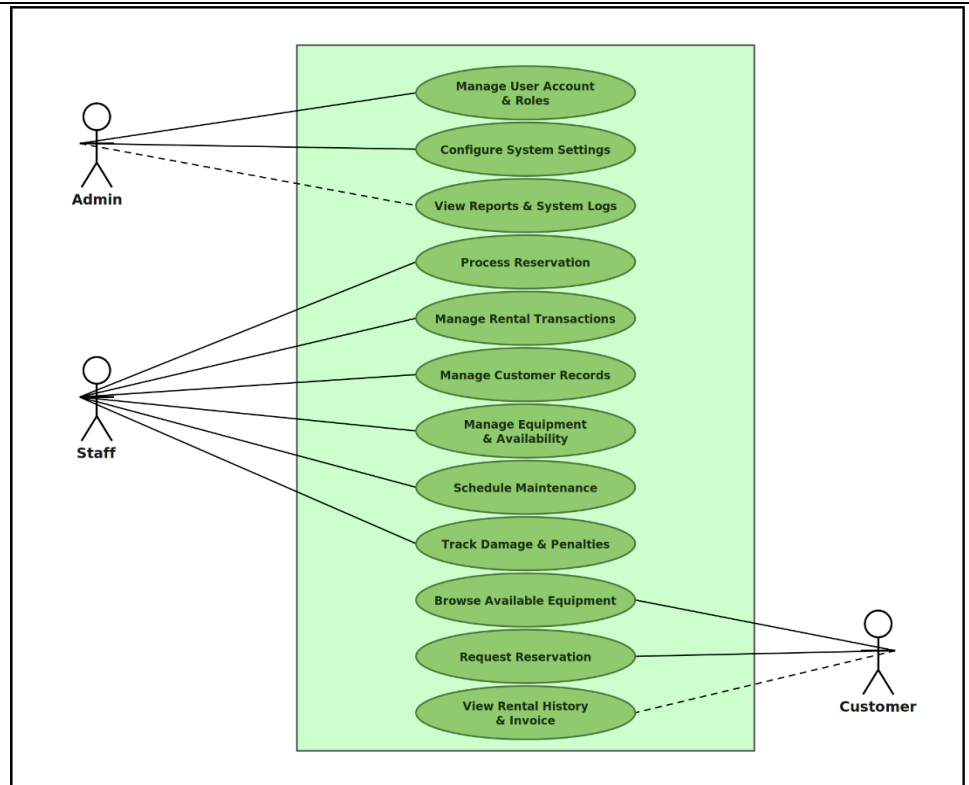


NAME	Angelo Ranes
PROJECT TITLE	RentalSphere: ERP and CRM System for Event & Party Equipment Rental Businesses
SUBJECT: CODE: TIME:	IT15/L Integrative Programming and Technologies 3481 8:00AM – 10:00AM
TOPIC (Type of Business Process)	#30 Rental Management Systems
Products/Services	Event & Party Equipment Rental Services
Tools Needed:	Visual Studio Code – IDE, SQL Server Management Studio (SSMS), Domain, Hosting (TBD)
Technology:	ASP.NET Core MVC (C#), Entity Framework Core, Microsoft SQL Server, Bootstrap (Razor Views)
Security Features	Authentication and Authorization, Role-Based Access Control (RBAC), Input Validation, Logging and Monitoring.
Target user/s:	Admin, Staff, Customer
Subs System / Management Transaction/Modules	1. Equipment Management Module 2. Customer Management Module 3. Reservation Management Module 4. Rental Transactions Module 5. Equipment Availability Module 6. Maintenance Management Module 7. Billing Module 8. Customer CRM Module 9. Damage & Penalty Tracking Module 10. Reports Module

LOGO	 RentalSphere EVENT & EQUIPMENT RENTAL ERP/CRM
Project Objectives:	1. To design and develop a centralized web-based ERP and CRM system for event and party equipment rental businesses using ASP.NET Core MVC within the current term. 2. To automate reservation and equipment availability tracking, reducing double-booking and manual scheduling conflicts. 3. To implement a role-based billing and transaction module that accurately records rental fees, damage penalties, and payment status per reservation. 4. To provide management with real-time reports on equipment usage, maintenance status, and revenue through a centralized dashboard.
Project Description:	RentalSphere is proposed for small to mid-sized event and party equipment rental businesses that currently rely on manual logs, spreadsheets, or disconnected tools to manage bookings, inventory, and billing. Its target users include rental staff who process reservations and transactions, inventory/maintenance officers who track equipment condition and availability, administrators who oversee operations and reporting, and customers who can browse and reserve equipment online. By centralizing these processes, RentalSphere aims to reduce scheduling conflicts, improve equipment turnaround visibility, and give business owners clearer financial and operational reporting, ultimately supporting more reliable service for customers and less manual workload for staff. The system will be built on ASP.NET Core MVC using C#, with Entity Framework Core handling data access and Microsoft SQL Server serving as the database. The front end will use Razor Views with Bootstrap for a responsive interface, developed in Visual Studio 2022 / Visual Studio Code. This stack was chosen for its strong tooling support, built-in security features for authentication and role-based access, and scalability for handling multiple concurrent reservations and transactions, aligning with the course's required .NET technology stack.

Use Case Diagram (Role-Based Access)



RentalSphere Role-Based Access Diagram

Admin

- Manage user accounts and roles
- Configure system settings
- Access all modules across the system
- View system logs and reports

Staff

- Process reservations and rental transactions
- Manage customer records
- Record billing and payment status
- Manage equipment records and availability
- Schedule and log maintenance activities
- Flag and record damaged equipment

Customer

- Browse available equipment
- Request and view reservations
- View rental history and invoices
- Update personal profile

Date Submitted:

Teacher's Feedback

	Student's Signature (after feedbacking) Teacher's Signature